

Final Exam
Logic and Sets for AI

*This exam has **3 pages** and **6 exercises** with a total of **66 points**. The result will be computed as $1 + \left(\frac{\text{total}}{66} \cdot 9\right)$.*

Note that in this course, 0 is considered a natural number. If you are, in any exercise, unsure about the definition of some term/operator/... state your assumed definition at the beginning of the exercise.

*Please put **your name on every sheet** you hand in, and write as clearly as possible. Submissions (even parts) without a name might **not be considered**. If we cannot read a statement it will be considered **incorrect**.*

Good luck!

1. Equivalence classes for $\equiv$ (9 points points)

Group the following twelve formulas into their semantic equivalence classes:

- | | | |
|--|--|---|
| (1) p | (2) $p \rightarrow (\neg q \wedge \neg p)$ | (3) $\neg p \rightarrow (\neg q \wedge \neg p)$ |
| (4) $\neg p \rightarrow (q \wedge p)$ | (5) $p \rightarrow (\neg q \wedge p)$ | (6) $\neg p \rightarrow (\neg q \wedge p)$ |
| (7) $p \wedge (q \rightarrow \neg p)$ | (8) $\neg p \wedge (q \rightarrow \neg p)$ | (9) $p \wedge (q \rightarrow \neg p)$ |
| (10) $\neg p \wedge (q \rightarrow p)$ | (11) $p \wedge (\neg q \rightarrow p)$ | (12) $\neg p$ |

Solution: There are the following equivalence classes:

- | | |
|-------------------------|------------------------|
| (a) (1), (4), (6), (11) | p |
| (b) (12), (2), (8) | $\neg p$ |
| (c) (3) | $p \vee \neg q$ |
| (d) (5) | $\neg p \vee \neg q$ |
| (e) (7), (9) | $p \wedge \neg q$ |
| (f) (10) | $\neg p \wedge \neg q$ |

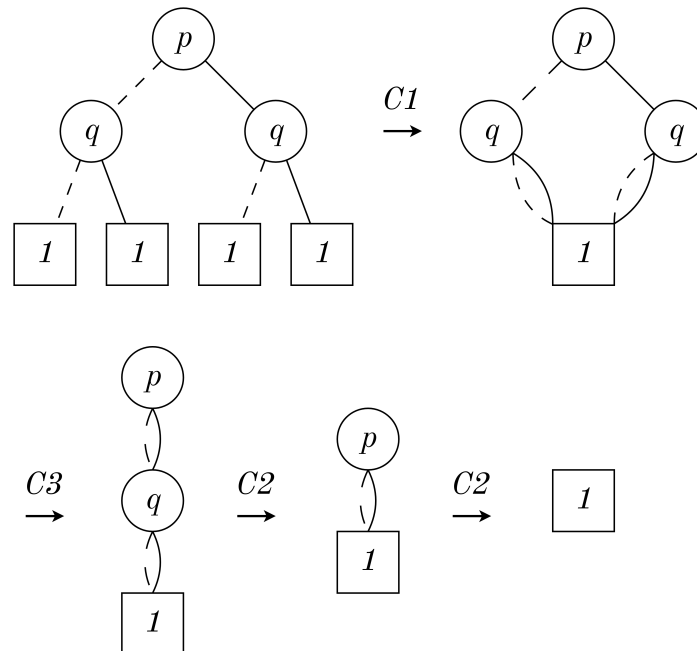
2. OBDDs/Logic circuits (5+4+5 points)

- (a) Draw the binary decision tree for the following formula. Use the order p, q, r for the variables. Apply the reduction algorithm to the tree until you've arrived at a reduced OBDD. Make sure to name all the steps.

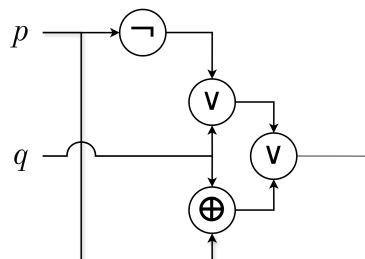
$$(p \rightarrow q) \vee (q \oplus p)$$

- (b) Draw the circuit diagram for the formula.
- (c) Draw the OBDDs for both disjuncts of the equation separately, denoted as O_1 and O_2 . Use them to construct the OBDD for $O_1 \wedge O_2$.

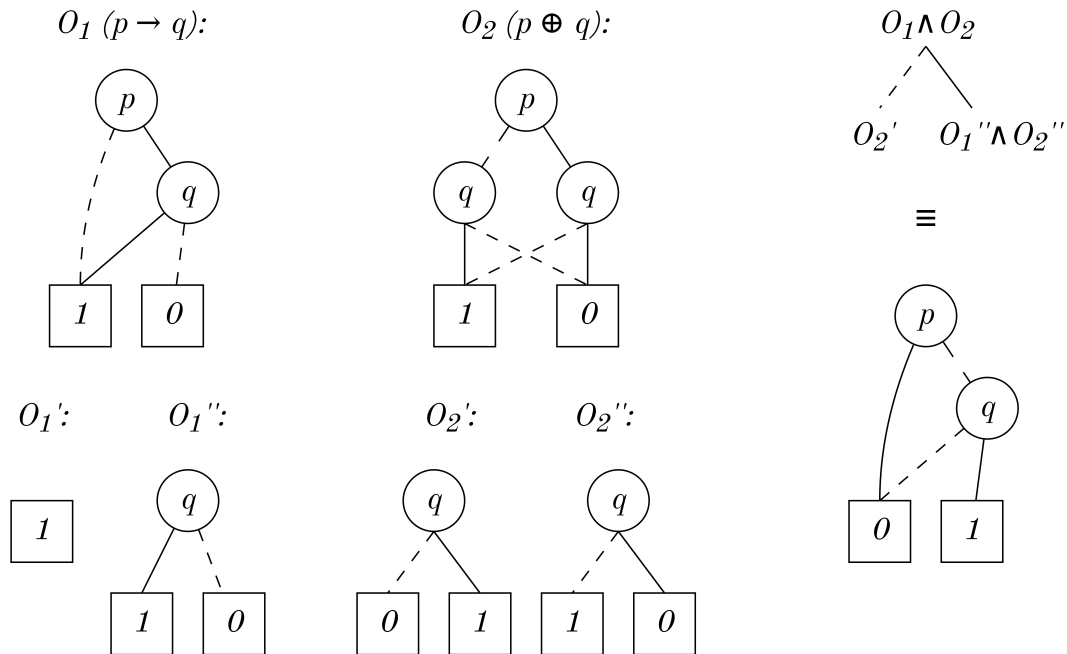
Solution: (a) The tree can be constructed without the variable r , which does not figure in the formula. Every transition should apply one rule (possibly multiple times) and that rule should be named. The final result must be the single leaf node '1'.



- (b) There are many possible approaches to constructing the circuit diagram in this case. Student may use the XOR gate if they so wish, since no restrictions were specified. Alternatively, they may decompose it to AND, OR and NOT connectors. However, an IMPL gate used in the same manner as all the other previously introduced binary gate **will result in an ambiguous diagram** and an incomplete circuit - the implication must be decomposed.



- (c) Construct two OBDDs for the separate conjuncts ($p \text{ IMPL } q$, $p \text{ XOR } q$). Note the exercise doesn't specify the trees must be reduced - even though the ones provided below are, an un-pruned tree is also a valid answer.



Students must i) construct both trees individually, and ii) indicate that they used them to construct the third tree, and not merely derived it from a truth table or by hand. This can take the form of writing out the formula for conjunctions of trees, writing out the sub-trees with their appropriate apostrophe names, or documenting the entire recursive process.

3. Predicate Logic (2+8 points)

- (a) Using the following predicates and variables represent the sentence *"If the Earth orbits the Sun, it is a planet."*

$O(x,y) := x$ orbits around y

$P(x) := x$ is a planet

$e :=$ The Earth

$s :=$ The Sun

- (b) Add one predicate to the above. Use all three predicates and any number of quantifiers to represent the following sentences:

(i) *"Some planets have a moon orbiting around them."*

(ii) *"Everything that orbits around the sun is either a planet or a moon."*

Note that a planet cannot be a moon.

Solution: (a) $O(e, s) \rightarrow P(e)$

(b) $M(x) := x$ is a moon

(i) $\exists x \exists y (P(x) \wedge M(y) \wedge O(y, x) \wedge \neg M(x))$

(ii) $\forall x (O(x, s) \rightarrow P(x) \oplus M(x))$

4. Equivalence Relations (3+6 points)

The set $\mathbb{Z} \setminus \{0\}$ (non-zero integers) can be partitioned into the following four sets: the set of positive even, positive odd, negative even, and negative odd numbers. Let R be the equivalence relation in $\mathbb{Z} \setminus \{0\}$ that has these four sets as its equivalence classes.

- (a) Give a complete system of representatives for R .

Solution: $S = \{-2, -1, 1, 2\}$

- (b) Consider the relation S on $\mathbb{N}$ that is defined by

$$xSy \iff \text{sum of digits in } x = \text{sum of digits in } y$$

.

For example, $14 S 212$, as the sum of digits in both numbers is 5. Show that S is an equivalence relation.

Solution: For S to be an equivalence relation, it has to be reflexive, transitive, and symmetric.

- S is reflexive as every number has the same sum of digits as itself.
- S is symmetric, as if x has the same sum of digits as y , it is clear that y has the same sum of digits as x .
- S is transitive, as if x has the same sum of digits as y , and y has the same sum of digits as z , then it is clear that x has the same sum of digits as z .

5. Functions (6+3+5 points)

Consider the functions $sqr : \mathbb{R} \rightarrow \mathbb{R}$, $add_2 : \mathbb{R} \rightarrow \mathbb{R}$, and $mul_2 : \mathbb{R} \rightarrow \mathbb{R}$, which are defined as follows:

$$sqr(x) := x^2, \quad add_2(x) := x + 2, \quad mul_2(x) := x * 2$$

- (a) Give a description (i.e., formula for $f(x)$) for the function f defined below and provide its domain of definition and range.

$$f := mul_2^{-1} \circ sqr^{-1} \circ add_2$$

Solution: $\frac{\sqrt{x+2}}{2}$. The domain of definition is $D = \{x : x \geq -2\}$, and the range is $R = \{y : y \geq 0\}$.

- (b) Give the inverse of f as a composition of sqr , add_2 , and mul_2 .

Solution: $add_2^{-1} \circ sqr \circ mul_2$

- (c) Now consider the function $abs : \mathbb{R} \rightarrow \mathbb{R}$, where $abs(x) = |x|$ denotes the absolute value of x . Additionally, the function g is defined as $g(x) = abs \circ mul_2$.

Is the function g total, injective, and/or surjective?

Solution: The domain of g is $\mathbb{R}$, which is also its domain of definition. Therefore, it is total. It is not injective, as $x_1 \neq x_2$ does not imply that $g(x_1) \neq g(x_2)$ (for example, $g(-1) = g(1) = 2$). It is not surjective since the range is not identical to the codomain, as the former is only the non-negative numbers in $\mathbb{R}$, while the latter is $\mathbb{R}$.

6. Induction (2+8 points)

We claim that the following statement is true for all natural numbers $n \geq 1$.

$$\sum_{k=1}^n k * (k+1) = \frac{n * (n+1) * (n+2)}{3}$$

(a) Verify by explicit computation that the claim is true for $n = 1$ and $n = 2$.

Solution: In case of $n = 1$: $1 * (1+1) = 2$ and $\frac{1*(1+1)*(1+2)}{3} = 2$. In case of $n = 2$:
 $1 * (1+1) + 2 * (2+1) = 8$ and $\frac{2*(2+1)*(2+2)}{3} = 8$.

(b) Prove by mathematical induction that the statement holds true for all natural numbers $n \geq 1$.

Solution: The steps of mathematical induction are as follows:

- Base case ($n = 1$): as proven above.
- Inductive step: let $m \geq 1$. Let us assume that $\sum_{k=1}^m k * (k+1) = \frac{m*(m+1)*(m+2)}{3}$ (Induction Hypothesis). Then, on the one hand:

$$\begin{aligned} & \sum_{k=1}^{m+1} k * (k+1) \\ &= \sum_{k=1}^m k * (k+1) + (m+1)(m+1+1) \\ &= \frac{m * (m+1) * (m+2)}{3} + (m+1)(m+2) \quad \text{IH} \\ &= \frac{m * (m+1) * (m+2)}{3} + \frac{3 * (m+1)(m+2)}{3} \\ &= \frac{(m+1) * (m+2) * (m+3)}{3} \end{aligned}$$

And on the other hand:

$$\begin{aligned} & \frac{(m+1) * (m+1+1) * (m+1+2)}{3} \\ &= \frac{(m+1) * (m+2) * (m+3)}{3} \end{aligned}$$

- Therefore, $\sum_{k=1}^m k*(k+1) = \frac{m*(m+1)*(m+2)}{3}$ implies $\sum_{k=1}^{m+1} k*(k+1) = \frac{(m+1)*(m+1+1)*(m+1+2)}{3}$ for all natural number $m \geq 1$, which completes the proof.

End of the exam!